

YoonwooJeong



Contact

jjw123822@gmail.com

Languages

Korean (native),
English (fluent)

Skills

Python, PyTorch,
CUDA, C/C++

GitHub

github.com/jeongyw12382

Website

jeongyw12382.github.io

Google Scholar

yunwoojeong.scholar

I am a computer vision researcher at the Computer Vision Lab, POSTECH, advised by Prof. Minsu Cho. My research centers on **integrating 3D geometric understanding into vision foundation models** — spanning neural radiance fields, multi-view reasoning, and visual perception in large multimodal models. I am broadly interested in building spatially-aware foundation models that can perceive, reason about, and interact with the 3D world.

Education

2021–2027	Pohang University of Science and Technology (POSTECH) Integrated M.S./Ph.D. student, Graduate School of Artificial Intelligence	Pohang, Korea
2017–2020	Pohang University of Science and Technology (POSTECH) B.S. in Computer Science and Engineering	Pohang, Korea
2014–2017	Korea Science Academy of KAIST	Busan, Korea

Experience

2026	Research Intern, Meta DINO Team [Upcoming] Under the supervision of Maxime Oquab.	Paris, France
2025	Research Intern, NVIDIA Research Developed 3D foundation models under the supervision of Jaesung Choe.	Taipei, Taiwan
2022–2023	Research Intern, Kakao Brain Developed text-to-3D diffusion models under the supervision of Doyup Lee.	Pangyo, Korea
2018–2020	Research Intern, Computer Vision Lab, POSTECH Developed robust visual correspondence models under the supervision of Jongmin Lee.	Pohang, Korea
2017–2018	Research Intern, Data Intelligence Lab, POSTECH Developed image style transfer models.	Pohang, Korea

Publications

* denotes work currently under review.

arXiv 2026*	Cog3DMap: Multi-View Vision-Language Reasoning with 3D Cognitive Maps Constructing 3D cognitive map to enhance 3D reasoning of VLMs.	co-1st author
arXiv 2026*	Vision-aligned Latent Reasoning for Multi-modal Large Language Model Injecting knowledge from visual foundation models into multi-view VLM.	2nd author
NVIDIA-Tech*	MV-SAM: Multi-view Promptable Segmentation using Pointmap Guidance Extended SAM to a multi-view setting without 3D or video datasets.	1st author
NVIDIA-Tech*	Quantile Rendering: Efficiently Embedding High-dimensional Features on 3D Gaussian Splatting A generalizable 3D-GS network for open-vocabulary segmentation.	1st author
arXiv 2025	RoDyGS: Robust Dynamic Gaussian Splatting for Casual Videos Pose-free dynamic neural fields leveraging geometric foundation models.	1st author
ECCV 2024	NVS-Adapter: Plug-and-Play Novel View Synthesis from a Single Image A parameter-efficient adapter for text-to-3D generative models.	1st author

ICML 2023	Stable and Consistent Prediction of 3D Characteristic Orientation via Invariant Residual Learning A 3D canonicalization framework for robust part-level segmentation.	3rd author
NeurIPS-W 2022	SeLCA: Self-supervised Learning of Canonical Axes An image canonicalization framework improving matching robustness.	2nd author
NeurIPS-D 2022	PeRFception: Perception using Radiance Fields A large-scale neural field dataset for perception tasks.	1st author
CVPR 2022	Fast Point Transformer An efficient voxel-based architecture for 3D perception.	2nd author
BMVC 2021	Self-supervised Learning of Image Scale and Orientation Estimation A self-supervised framework for image canonicalization and matching.	2nd author
ICCV 2021	Self-Calibrating Neural Radiance Fields The first pose-free NeRF optimization pipeline.	1st author
WACV 2021	Learning to Distill Convolutional Features into Compact Local Descriptors A memory-efficient feature extractor for geometric matching.	2nd author

Projects

* updated on 26/04/17.

2022	NeRF-Factory: An Awesome PyTorch NeRF Library [★: 1.3K] A PyTorch reimplementation of state-of-the-art NeRF models. Link: github.com/kakaobrain/nerf-factory	Project Leader
2022	PeRFception: Perception using Radiance Fields [★: 0.3K] A large-scale dataset for perception tasks based on neural fields. Link: github.com/POSTECH-CVLab/PeRFception	Project Leader
2021	SCNeRF: Self-calibrating Neural Radiance Fields [★: 0.5K] Neural radiance fields for uncalibrated images. Link: https://github.com/POSTECH-CVLab/SCNeRF	Project Leader

Awards

2025	Postechian Fellowship Recognized for outstanding innovation and research excellence.	
2023	Postechian Fellowship Awarded for exceptional leadership and community engagement.	
2023	POSTECH Research Award Selected paper: <i>PeRFception</i> .	Excellence Award
2023	Samsung Human Tech Paper Award Selected paper: <i>PeRFception</i> .	Encouragement Award
2022	Qualcomm Innovation Fellowship Korea Two papers selected: <i>Fast Point Transformer</i> and <i>SC-NeRF</i> .	
2022	Samsung Human Tech Paper Award Selected paper: <i>Fast Point Transformer</i> .	Silver Award
2022	POSTECH Research Award Selected paper: <i>Self-Calibrating Neural Radiance Fields</i> .	Top Award
2021	POSTECH Research Award Selected paper: <i>Self-supervised Learning of Image Scale and Orientation</i> .	Top Award
2019	PPC – POSTECH Programming Contest	Top Award

Presentations

2025	RLWRDL Presented <i>RoDyGS</i> , a pose-free dynamic neural field framework.	Seoul, Korea
2024	Korea AI Summit Presented <i>NVS-Adapter</i> , a text-to-3D diffusion model.	Seoul, Korea
2022	Guest Lecture – Kakao Brain, Kakao Enterprise, POSTECH, and MODULabs Delivered a lecture on recent advances in neural rendering.	Seoul, Korea
2021	Guest Lecture – Hanyang Erica SW/ICT Tech Competition and POSTECH Shared research experiences and insights from M.S. studies.	Online

Reviewing Services

2026	ICLR, CVPR, ECCV
2025	CVPR, NeurIPS, TPAMI
2024	ECCV, NeurIPS-D (Dataset and Benchmark Track)
2022	CVPR, ECCV, NeurIPS-D (Dataset and Benchmark Track)
2021	WACV
2020	NeurIPS

Extracurricular Activities

2023–2024	POSTEnglish – POSTECH English Debate Team
2023–2025	Climbmax – POSTECH Sports Climbing Team President of the club.
2023–2024	President, Student Council of GSAI Served as student president of the Graduate School of Artificial Intelligence, POSTECH.
2017–2018	POSCAT – POSTECH Computing Algorithms Team Participated in ACM-ICPC Regionals.
2017–2019	CHEERO – POSTECH Cheerleading Club President of the club.